Soham Karpe, M. Eng.

AI Systems Engineer

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Steinweg 17, 93413 Cham, Germany | Willing to relocate

Date of Birth: 03.02.1998, India Driving license: Class B Portfolio

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PROFILE

Experienced AI systems engineer in the field of image processing and industrial vision, with hands-on experience in developing real-time inspection systems for industrial production environments. Experienced in image processing algorithms, camera calibration, and the integration of industrial cameras and sensor systems. Excellent knowledge of C++, Python, and OpenCV. Strong interest in the further development of image quality, algorithms, and sensor-based systems, particularly in the field of automated inspection.

PROFESSIONAL EXPERIENCE

AI Systems Engineer, mech-tron GmbH & Co. KG, Roding, Germany Jul 2024 – Present

- Implementation of standard image processing algorithms for feature extraction and defect detection: filtering, thresholding (Otsu), edge detection (Canny, Sobel), blob analysis, morphological operations (erosion, dilation), template matching, Hough transform, and histogram analysis
- Development of software solutions using C, C++, Python, and OpenCV
- Integration of industrial cameras and sensor systems, distortion correction, and camera calibration including geometric calibration and pixel-to-mm conversion
- Optimization of image quality for reliable inspection results
- Development and integration of AI-based inspection systems using YOLO and deep learning models combined with classical image processing
- Creation and preparation of customised datasets for AI models (annotation, training)
- Development of HMI interfaces with Qt for production systems
- Deployment of inspection solutions on embedded and edge systems (Raspberry Pi, IoT)

Junior Software Developer, Mar 2024 – Jun 2024
mech-tron GmbH & Co. KG, Roding, Germany

- Development of computer vision pipelines using OpenCV and deep learning for real-time object recognition
- Deployment and optimization of AI models on edge systems, including performance analysis

Software Engineer, Evoka Technologies, Pune, India Jul 2020 – Jul 2022

- Development, testing, and debugging of software modules in C++ and Python for prototypes in early development phases, as well as support for system architecture and technical implementation
- Participation in hardware-software integration and system validation in collaboration with interdisciplinary teams, including system design, code reviews, and test strategies
- Contribution to end-to-end product development from requirements analysis through implementation and commissioning, including continuous optimization and technical decision-making in a dynamic environment

Intern and Bachelor-thesis, Sep 2019 – Jun 2020
Profiroll Technologies India Pvt. Ltd., Pune, India

- Designed and optimized a compact hydraulic straightening machine to improve mechanical alignment and efficiency.
- Performed system modeling and engineering analysis to evaluate mechanical performance and structural stability.

EDUCATION

M.Eng. Artificial Intelligence for Smart Sensors and Actuators Oct 2022 – Mar 2026 | Cham, Germany
Technische Hochschule Deggendorf (Deggendorf Institute of Technology- DIT)

Artificial Intelligence, Model Based Function Engineering, Machine Learning and Deep Learning, Robotics and Autonomous Systems, Big Data, Computer Vision, System Design and Intercommunication, Embedded Systems.

Master's Thesis: mech-tron GmbH & Co. KG, Roding (Nominated for Best Thesis Award)

Design and Development of a Real-Time AI-Based Visual Inspection System
Embedded System Integration • Industrial Software Architecture • Computer Vision & Image Processing Pipelines • Industrial Camera Integration • Qt-Based HMI/GUI Development

Bachelor of Engineering, Pune University, India — Jul 2016 – Jun 2020
Programming Fundamentals, Engineering Mathematics, Data Structures (Fundamentals), Control Systems, Mechatronics Systems, FEA, Applied Mechanics, Strength of Materials, Dynamics of Machinery

PROJECT

Liveness detection using computer vision (Bildverarbeitung) — Mar 2024 – Jun 2024
- Developed a YOLOv8-based object recognition system achieving 91% accuracy across a multi-camera setup, with a complete real-time inference pipeline optimized for edge platforms and validated under varying lighting conditions and camera angles.

Automated Testing & Multi-Camera Integration Framework for Computer Vision Systems — Oct 2024 – Feb 2025
- Developed a pytest-based automated testing framework for validating YOLOv8 models, including CI/CD pipeline integration for continuous model testing and performance benchmarking.
- Integrated multiple industrial cameras for synchronized image capture and real-time processing, implementing camera calibration, sensor fusion, and spatial alignment for 3D scene understanding.

Industrial Inspection HMI with Built-in Image Labelling and Annotation — Jun 2025 – Aug 2025
- Designed and developed a Qt-based HMI for real-time machine vision inspection, featuring live camera feed, defect visualization, status monitoring, and configurable inspection parameters.
- Built an integrated image labelling and annotation module within the HMI, enabling operators to capture, tag, and export defect images directly on the production line — eliminating external annotation tools and accelerating AI model retraining.

KNOWLEDGE AND TECHNICAL SKILLS

Programming languages
C, C++, C#, Python.

Test automation frameworks:
pytest, unittest, GitLab CI/CD, GitHub Actions Testing: Unit testing, integration testing, system testing Debugging & validation: Performance testing, test reports

Productivity Tools & Visualization
Visual Studio, Visual Studio Code, Git/GitHub, CI/CD pipelines, Linux, Windows, Bash scripting, PowerShell, QEMU

Image Processing & Computer Vision
OpenCV (C++/Python), image processing algorithms (filtering, reconstruction, calibration, artifact removal, compression), object detection (YOLOv8), classical image processing methods, real-time inference optimization with GPU, image quality optimization, learning-based vision integration

Embedded Systems & Hardware Integration
Multi-camera integration (machine vision cameras), embedded platforms (Raspberry Pi, ESP32, NVIDIA Jetson, Arduino Mega), edge inference, sensor integration, hardware-software integration

Languages
Deutsch — B1
Englisch — B2
Hindi & Marathi — Native

AI & Machine Learning
PyTorch, TensorFlow, Model Training & Optimization, Edge Deployment, TensorRT (basics), Real-Time ML Inference, Halcon

Soft Skills
Analytical thinking, problem-solving, communication, teamwork, initiative, goal-oriented

CERTIFICATIONS

Machine Learning with Python • Introduction to Deep Learning & Neural Networks with Keras • AWS Certified Machine Learning - Specialty • AWS AI Practitioner Certification Prep

INTERESTS

Experienced in developing and refactoring real-time image processing systems using C, C++, and Python (AI/ML), with a focus on calibration, image filtering, error correction, compression, and sensor quality analysis — delivering reliable image quality optimization for quality-critical inspection applications in the food, pharmaceutical, metal, packaging, and life sciences industries.